

FOUNDATIONAL MATHEMATICS FOR MACHINE LEARNING

PROJECT REPORT

Stock Portfolio Optimization

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Introduction & Objectives

This report presents a complete quantitative analysis of a cross-sector, cross-geography equity portfolio. Daily adjusted closing prices of eleven stocks were collected and transformed into log-returns. The resulting dataset was used to analyse return behaviour, volatility, and inter-asset relationships through covariance and correlation analysis. These statistical measures form the foundation for Principal Component Analysis (PCA) and mean-variance portfolio optimization.

The report covers the following steps:

1. **Exploratory analysis** of price trends, return distributions, rolling volatility, and risk-adjusted performance.
2. **Covariance and correlation analysis** to quantify pairwise asset relationships.
3. **PCA** to decompose the covariance structure into latent market risk factors.
4. **Mean-variance optimisation** via the Lagrangian closed-form solution to trace the efficient frontier.
5. **PCA factor-model comparison** against the full-covariance frontier to assess the cost of dimensionality reduction.

Key mathematical Concepts

Log Returns

For an asset with price P_t , the daily log-return is defined as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right).$$

Log returns are time-additive, statistically more stable than price levels, and simplify multi-period portfolio analysis.

Volatility

Annualised volatility of asset i is

$$\sigma_i^{\text{ann}} = \sigma_i^{\text{daily}} \sqrt{252},$$

where 252 is the standard number of trading days per year. Higher volatility indicates greater uncertainty in future returns.

Correlation and Diversification

The Pearson correlation coefficient between assets i and j is

$$\rho_{ij} = \frac{\text{Cov}(r_i, r_j)}{\sigma_i \sigma_j}, \quad -1 \leq \rho_{ij} \leq 1.$$

Combining assets with lower correlations reduces overall portfolio risk.

Data Collection and Cleaning

Historical adjusted closing prices were obtained using the `yfinance` library. Eleven stocks were selected from five sectors to ensure cross-industry diversification.

The selected assets are: AAPL, MSFT, GOOGL (Technology); JPM, BAC (Banking); XOM, CVX (Energy); JNJ, PFE (Healthcare); TCS.NS, INFY.NS (Indian IT).

The sample period is **January 2018 – January 2024** (1 318 daily observations). Daily log-returns were computed as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right),$$

where P_t denotes the adjusted closing price on day t .

Normalised Price Trends

To compare long-term performance on a common scale, all price series were normalised to a base value of 100 at the start of the sample period.

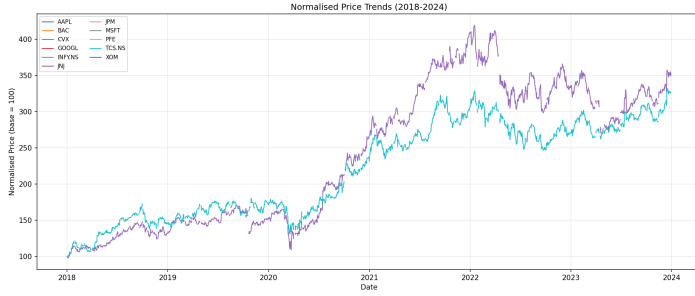


Figure 1: Normalised price trends (base = 100 at 2018-01-01).

Summary Statistics

Annualised expected return, volatility, and Sharpe ratio for each asset were estimated as

$$\mu_i^{\text{ann}} = 252 \cdot \bar{r}_i, \quad \sigma_i^{\text{ann}} = \sigma_i^{\text{daily}} \sqrt{252}, \quad S_i = \frac{\mu_i^{\text{ann}}}{\sigma_i^{\text{ann}}},$$

assuming a risk-free rate of zero.

2.6.1 Results

Table 1: Annualised summary statistics (2018–2024).

Asset	Ann. Return (%)	Ann. Volatility (%)	Sharpe Ratio
AAPL	23.91	31.87	0.750
MSFT	25.58	30.36	0.842
GOOGL	20.30	30.92	0.657
JPM	10.77	30.64	0.352
BAC	8.54	34.41	0.248
XOM	11.44	32.60	0.351
CVX	5.64	34.51	0.163
JNJ	2.77	20.35	0.136
PFE	-0.70	25.40	-0.028
TCS.NS	20.02	24.52	0.816
INFY.NS	26.45	26.59	0.995

Technology stocks (AAPL, MSFT, GOOGL) and Indian IT stocks (TCS.NS, INFY.NS) generated the highest annualised returns and Sharpe ratios. INFY.NS achieved the highest individual Sharpe ratio (0.995), followed closely by MSFT (0.842) and TCS.NS (0.816).

Banking stocks (BAC, JPM) and energy stocks (CVX, XOM) exhibited higher volatility due to macroeconomic and commodity-price sensitivity. Healthcare stocks (JNJ, PFE) displayed stable, low-volatility behaviour; PFE produced a slightly negative return over the period.

Return Distribution Analysis

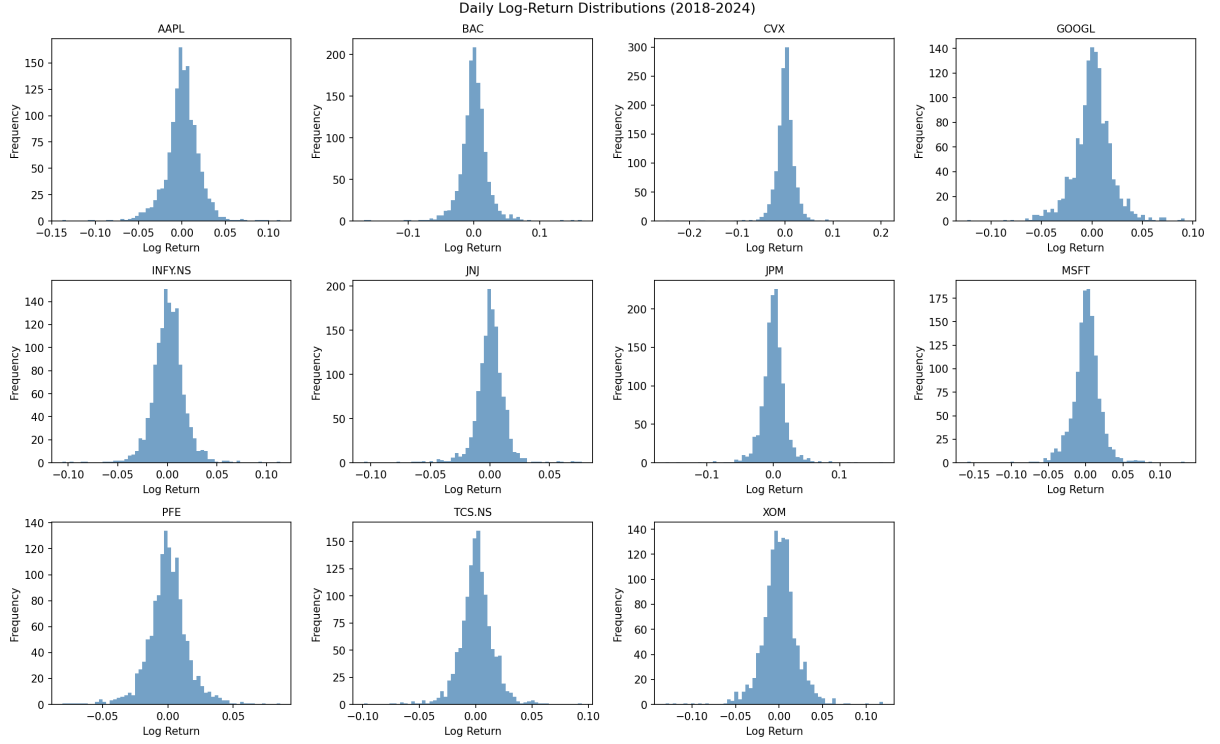


Figure 2: Daily log-return distributions (2018–2024).

2.7.1 Results

Figure 2 shows the distribution of daily log-returns for each asset. Most distributions are bell-shaped and centred near zero, consistent with small daily moves occurring more frequently than large ones. Several assets display heavier tails than a normal distribution, indicating elevated probability of extreme market events.

Technology and energy stocks exhibit wider distributions reflecting higher return variability. Healthcare stocks (JNJ, PFE) display narrower distributions, consistent with their lower annualised volatilities in Table 1.

Risk-Return Profile

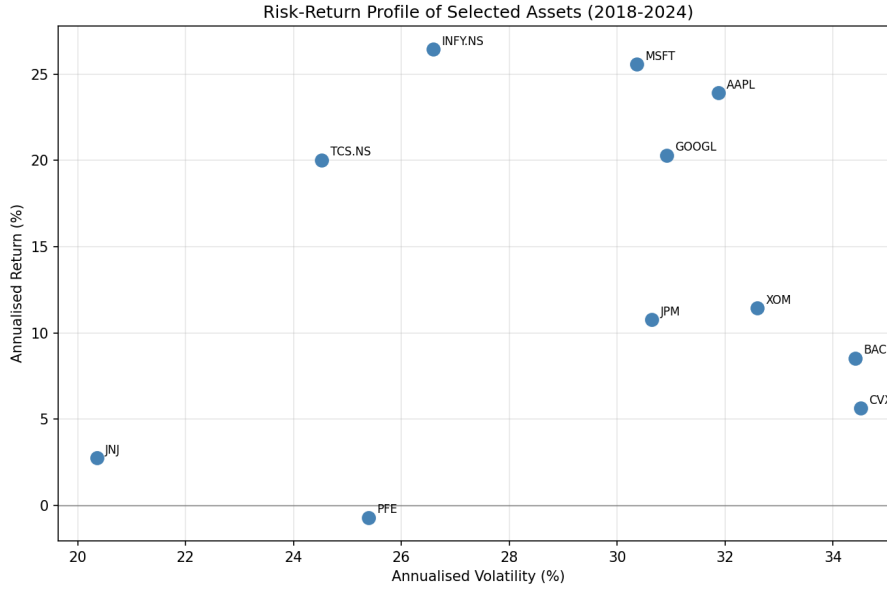


Figure 3: Risk–return profile of selected assets (annualised, 2018–2024).

2.8.1 Results

Figure 3 plots each asset’s annualised return against its annualised volatility. INFY.NS, MSFT, and TCS.NS occupy the most attractive region — relatively high return for moderate volatility. Banking and energy stocks cluster in a high-volatility, low-return region. JNJ and PFE occupy the low-volatility, low-return corner.

The scatter confirms the fundamental risk–return tradeoff and motivates the search for diversified portfolios that lie to the left of (i.e. dominate) individual assets.

Correlation Analysis

Diversification benefits arise when assets are imperfectly correlated. The Pearson correlation coefficient is

$$\rho_{ij} = \frac{\text{Cov}(r_i, r_j)}{\sigma_i \sigma_j}.$$

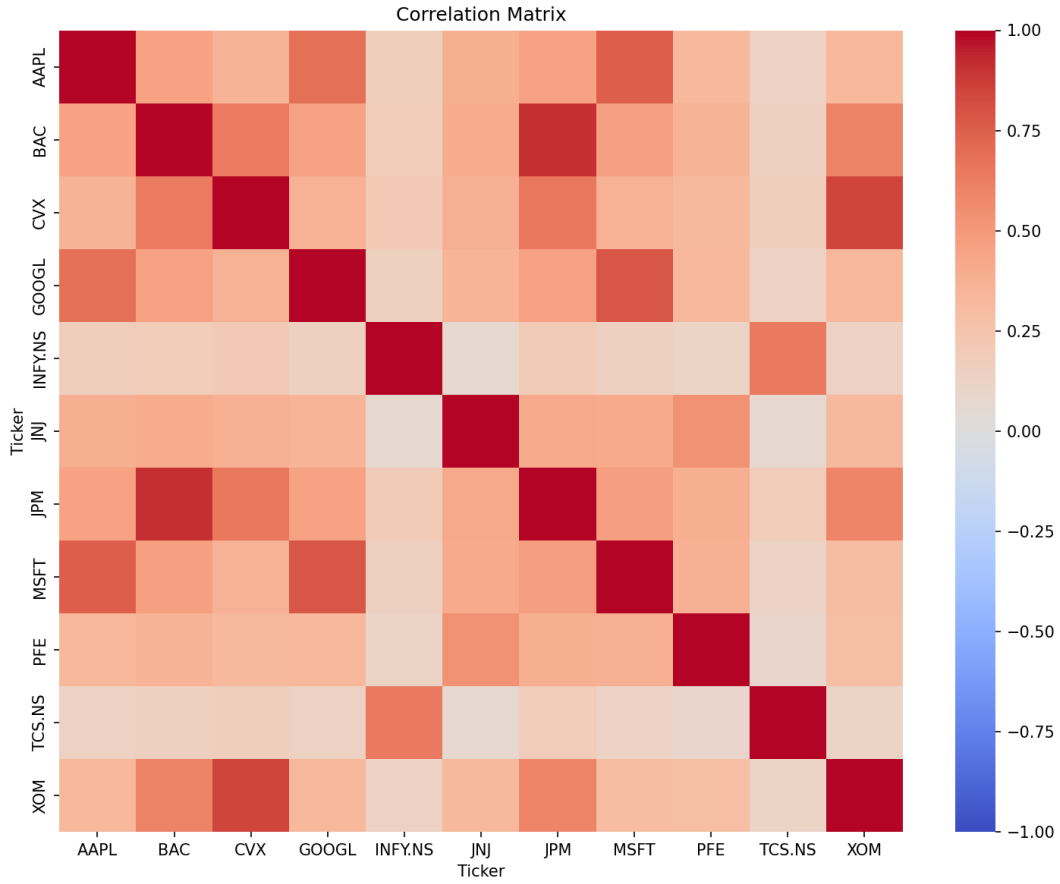


Figure 4: Correlation heatmap of daily log-returns (2018–2024).

2.9.1 Results

Figure 4 reveals clear sector-level clusters. BAC and JPM are strongly correlated (0.91), as are CVX and XOM (0.85), and the tech pair AAPL–MSFT (0.75) and GOOGL–MSFT (0.78). INFY.NS and TCS.NS share a correlation of ~ 0.65 due to common exposure to the Indian IT sector. Healthcare stocks (JNJ, PFE) show the weakest correlations with the rest of the universe, providing meaningful diversification potential.

Rolling Volatility Analysis

A 30-day rolling annualised volatility was computed as

$$\hat{\sigma}_t = \sqrt{252} \cdot \text{Std}(r_{t-29}, \dots, r_t).$$

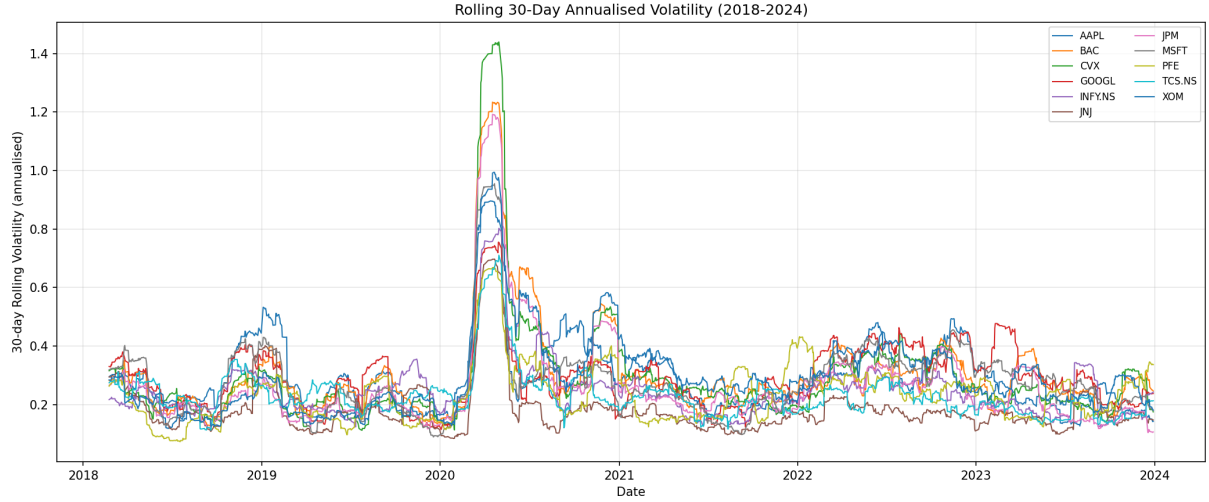


Figure 5: Rolling 30-day annualised volatility (2018–2024).

2.10.1 Results

Figure 5 shows that volatility is far from constant. All assets experienced sharp spikes during the COVID-19 market shock (early 2020), demonstrating that a single static covariance matrix may underestimate tail risk during stress periods. Banking and energy stocks showed the largest absolute volatility levels; healthcare stocks remained comparatively stable throughout.

Sharpe Ratio Analysis

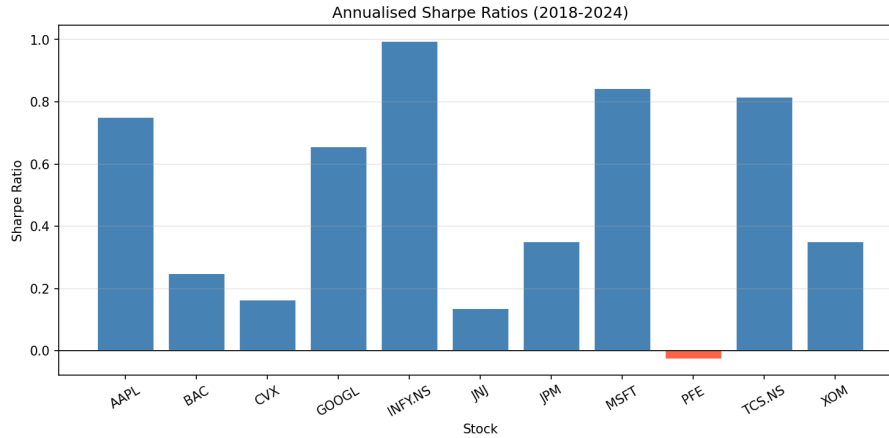


Figure 6: Annualised Sharpe ratios (2018–2024, $r_f = 0$).

2.11.1 Results

Figure 6 confirms that INFY.NS (0.995), MSFT (0.842), and TCS.NS (0.816) delivered the best risk-adjusted returns over the period. PFE is the only asset with a negative Sharpe ratio. Banking and energy stocks generated moderate positive Sharpe ratios despite their elevated volatility.

Key Findings from Exploratory Analysis

- Technology and Indian IT stocks exhibited the strongest long-term growth and highest individual Sharpe ratios.

- Banking and energy stocks experienced higher volatility due to sensitivity to macroeconomic and commodity market conditions.
- Healthcare stocks displayed stable, low-risk return characteristics.
- Strong correlations exist within sector groups (tech, banking, energy, Indian IT), while cross-sector correlations are weaker.
- Rolling volatility analysis confirmed that market risk is time-varying and can spike substantially during periods of stress.

Covariance and Correlation Analysis

Covariance Matrix

For a portfolio of n assets, the sample covariance matrix is

$$\hat{\Sigma} = \begin{bmatrix} \text{Cov}(r_1, r_1) & \cdots & \text{Cov}(r_1, r_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(r_n, r_1) & \cdots & \text{Cov}(r_n, r_n) \end{bmatrix}.$$

Diagonal entries are individual asset variances; off-diagonal entries measure pairwise co-movement.

4.1.1 Results

Table 2: Sample covariance matrix (daily, first three rows shown).

	AAPL	BAC	CVX	GOOGL	INFY.NS	JNJ	JPM	MSFT	PFE	TCS.NS	XOM
AAPL	0.000403	0.000198	0.000157	0.000266	0.000057	0.000099	0.000178	0.000290	0.000107	0.000044	0.000137
BAC	0.000198	0.000470	0.000299	0.000192	0.000067	0.000112	0.000381	0.000191	0.000125	0.000054	0.000270
CVX	0.000157	0.000299	0.000473	0.000159	0.000079	0.000105	0.000273	0.000154	0.000111	0.000057	0.000378

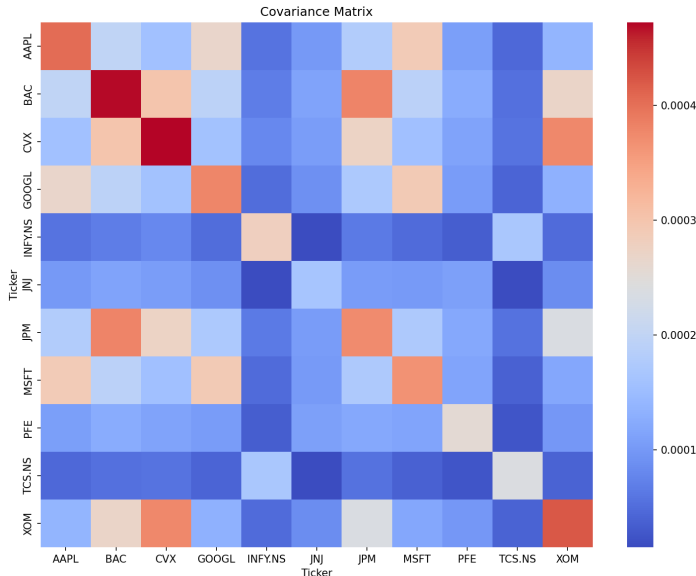


Figure 7: Covariance matrix heatmap.

Figure 7 shows the covariance heatmap. Warmer colours indicate stronger positive covariance.

BAC-JPM and CVX-XOM exhibit the largest off-diagonal covariances, reflecting tight within-sector co-movement. AAPL, MSFT, and GOOGL show notable pairwise covariances consistent with their high correlation. INFY.NS and TCS.NS covariances are smaller in absolute terms but elevated relative to cross-sector pairs.

No significant negative covariance entries are present, confirming that all assets tend to move in the same broad market direction.

Correlation Matrix

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \text{Corr}(X, Y) \leq 1.$$

4.2.1 Results

Table 3: Sample correlation matrix (first three rows shown).

	AAPL	BAC	CVX	GOOGL	INFY.NS	JNJ	JPM	MSFT	PFE	TCS.NS	XOM
AAPL	1.000	0.454	0.360	0.681	0.170	0.385	0.459	0.755	0.333	0.142	0.332
BAC	0.454	1.000	0.635	0.455	0.184	0.404	0.910	0.462	0.361	0.160	0.606
CVX	0.360	0.635	1.000	0.375	0.218	0.377	0.651	0.371	0.320	0.170	0.846

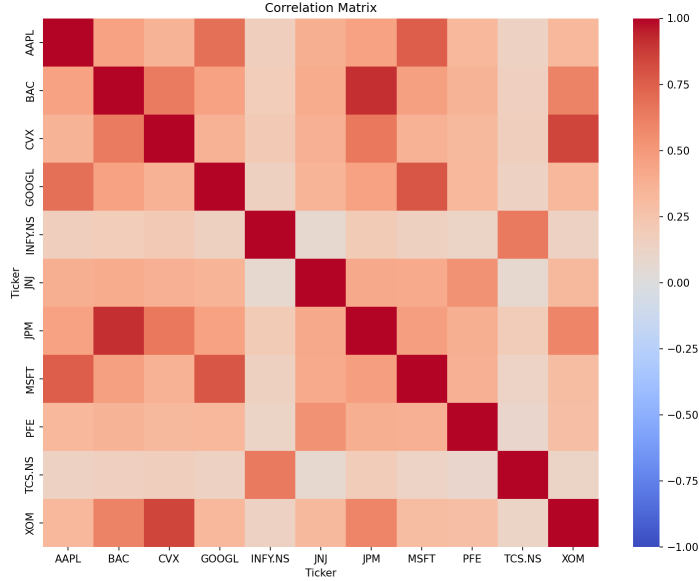


Figure 8 shows the correlation heatmap. Strong positive correlations: BAC–JPM (0.91), CVX–XOM (0.85), GOOGL–MSFT (0.78), AAPL–MSFT (0.75). INFY.NS–TCS.NS show a correlation of ~ 0.65 .

Compared with the raw covariance, the correlation matrix makes sector-level clustering visually clearer by removing scale differences across assets.

Figure 8: Correlation matrix heatmap.

Principal Component Analysis

PCA decomposes the covariance matrix via eigendecomposition:

$$\hat{\Sigma} = V\Lambda V^\top,$$

where V contains the eigenvectors (principal directions) and Λ the corresponding eigenvalues (variances along each direction).

Eigenvalues and Explained Variance

Table 4: Eigenvalues of principal components.

PC	Eigenvalue
1	0.001887
2	0.000566
3	0.000400
4	0.000261
5	0.000227
6	0.000127
7	0.000093
8	0.000090
9	0.000076
10	0.000061
11	0.000036

The first eigenvalue is substantially larger than the rest, indicating a dominant market-wide factor. The sharp scree after PC1 confirms that a small number of components captures most of the total variation.

Table 5: Explained and cumulative variance by PC.

PC	Expl. Var.	Cumulative
1	0.4934	0.4934
2	0.1481	0.6415
3	0.1047	0.7461
4	0.0683	0.8144
5	0.0594	0.8738
6	0.0332	0.9070
7	0.0243	0.9313
8	0.0234	0.9547
9	0.0198	0.9745
10	0.0161	0.9906
11	0.0094	1.0000

PC1 explains 49.3% of total variance. The first two components explain 64.2%, and the first three explain 74.6%. Cumulative variance exceeds 80% after four components and 87.4% after five.

This confirms that PCA provides an effective low-dimensional representation of portfolio risk.

PCA Factor Loadings

Table 6: Principal component loadings (top 3 PCs).

Stock	PC1	PC2	PC3
AAPL	-0.327	-0.449	-0.021
BAC	-0.424	+0.197	-0.070
CVX	-0.401	+0.394	-0.028
GOOGL	-0.318	-0.433	-0.039
INFY.NS	-0.109	+0.015	+0.738
JNJ	-0.160	-0.051	-0.060
JPM	-0.379	+0.170	-0.038
MSFT	-0.318	-0.462	-0.047
PFE	-0.187	-0.071	-0.023
TCS.NS	-0.088	+0.011	+0.656
XOM	-0.355	+0.407	-0.098

PC1 carries similar negative loadings across all stocks — a broad market (beta) factor.

PC2 separates energy and banking (positive: XOM, CVX, BAC, JPM) from technology (negative: AAPL, MSFT, GOOGL).

PC3 is dominated by INFY.NS (+0.738) and TCS.NS (+0.656), capturing the Indian IT sector factor.

5.2.1 Factor Loading Plot

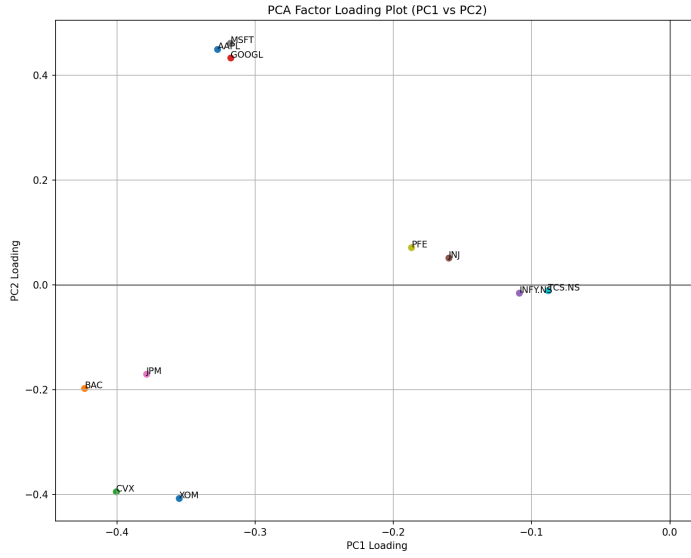


Figure 9 shows each stock's loading on PC1 and PC2.

Technology stocks (AAPL, MSFT, GOOGL) cluster in the lower-left: similar negative PC1 and PC2 loadings. Energy stocks (XOM, CVX) cluster upper-left: negative PC1 but positive PC2. Banking stocks (BAC, JPM) sit between tech and energy on PC2.

INFY.NS and TCS.NS are near the horizontal axis, as their dominant exposure is to PC3 (the Indian IT factor), not captured by this 2D view.

Figure 9: Factor loadings of stocks on PC1 and PC2.

Covariance Reconstruction Using PCA

The rank- k PCA approximation is

$$\hat{\Sigma}_k = V_k \Lambda_k V_k^\top,$$

where $V_k \in \mathbb{R}^{p \times k}$ and $\Lambda_k = \text{diag}(\lambda_1, \dots, \lambda_k)$.

5.3.1 Reconstructed Covariance Matrix

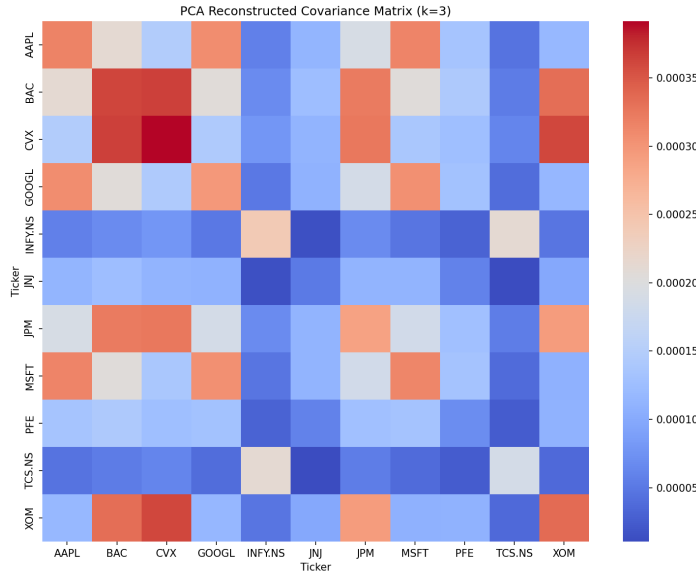


Figure 10 shows the covariance matrix reconstructed using $k = 3$ principal components.

The overall structure closely mirrors the full matrix. BAC–JPM and CVX–XOM relationships remain clearly visible, confirming that three dominant factors capture the majority of systematic market co-movement.

Figure 10: Covariance matrix reconstructed using $k = 3$ PCA components.

5.3.2 Reconstruction Error

The quality of the approximation is measured by the Frobenius norm:

$$\|\hat{\Sigma} - \hat{\Sigma}_k\|_F = 0.000404 \quad (k = 3).$$

This error is small relative to the covariance magnitudes ($\sim 10^{-4}$), confirming that three factors preserve the essential covariance structure.

Table 7: Sample reconstruction error matrix (first three rows, $k = 3$).

	AAPL	BAC	CVX	GOOGL	INFY.NS	JNJ	JPM	MSFT	PFE	TCS.NS	XOM
AAPL	8.63×10^{-5}	-1.43×10^{-5}	9.87×10^{-6}	-4.09×10^{-5}	-3.09×10^{-7}	-1.34×10^{-5}	-1.33×10^{-5}	-2.47×10^{-5}	-2.71×10^{-5}	-2.14×10^{-6}	2.03×10^{-5}
BAC	-1.43×10^{-5}	1.07×10^{-4}	-6.60×10^{-5}	-1.47×10^{-5}	-1.32×10^{-6}	-1.18×10^{-5}	5.78×10^{-5}	-1.28×10^{-5}	-1.72×10^{-5}	2.61×10^{-7}	-6.25×10^{-5}
CVX	9.87×10^{-6}	-6.60×10^{-5}	8.10×10^{-5}	1.45×10^{-5}	1.78×10^{-6}	-4.98×10^{-6}	-5.17×10^{-5}	1.62×10^{-5}	-1.46×10^{-5}	-4.84×10^{-6}	1.67×10^{-5}

The errors are small and centred around zero, confirming that the three-factor PCA model preserves the essential covariance structure with minimal information loss.

Mean-Variance Optimisation

Lagrangian Formulation

Given $\hat{\Sigma} \in \mathbb{R}^{p \times p}$, $\hat{\mu} \in \mathbb{R}^p$, and a scalar target return r^* , the minimum-variance portfolio solves

$$\min_{\mathbf{w}} \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^\top \hat{\mu} = r^*, \quad \mathbf{1}^\top \mathbf{w} = 1. \quad (1)$$

KKT conditions yield the **closed-form solution**

$$\mathbf{w}_{\text{opt}} = \hat{\Sigma}^{-1}(\lambda \hat{\mu} + \nu \mathbf{1}), \quad (2)$$

where

$$\lambda = \frac{Cr^* - B}{D}, \quad \nu = \frac{A - Br^*}{D}, \quad (3)$$

and

$$A = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}, \quad B = \hat{\mu}^\top \hat{\Sigma}^{-1} \mathbf{1}, \quad C = \mathbf{1}^\top \hat{\Sigma}^{-1} \mathbf{1}, \quad D = AC - B^2. \quad (4)$$

Minimum-Variance Portfolio

$$\mathbf{w}_{\text{min}} = \frac{\hat{\Sigma}^{-1} \mathbf{1}}{C}, \quad r_{\text{min}} = \frac{B}{C}, \quad \sigma_{\text{min}} = \frac{1}{\sqrt{C}}. \quad (5)$$

Maximum-Sharpe Portfolio

$$\text{SR}(\mathbf{w}) = \frac{\mathbf{w}^\top \hat{\mu}}{\sqrt{\mathbf{w}^\top \hat{\Sigma} \mathbf{w}}}. \quad (6)$$

Maximising (6) subject to $\mathbf{1}^\top \mathbf{w} = 1$ (short-selling allowed) is solved numerically via SLSQP constrained optimisation.

Efficient Frontier

Scalar Constants

Table 8 reports the Lagrangian constants computed from the annualised $\hat{\mu}$ and $\hat{\Sigma}$.

Table 8: Lagrangian scalar constants (annualised, 2018–2024).

A	B	C	$D = AC - B^2$	$\sigma_{\text{min}} = 1/\sqrt{C}$
2.141	5.551	43.850	63.074	0.151

Frontier Plot

Figure 11 shows the efficient frontier overlaid with all eleven individual assets and the two special portfolios.

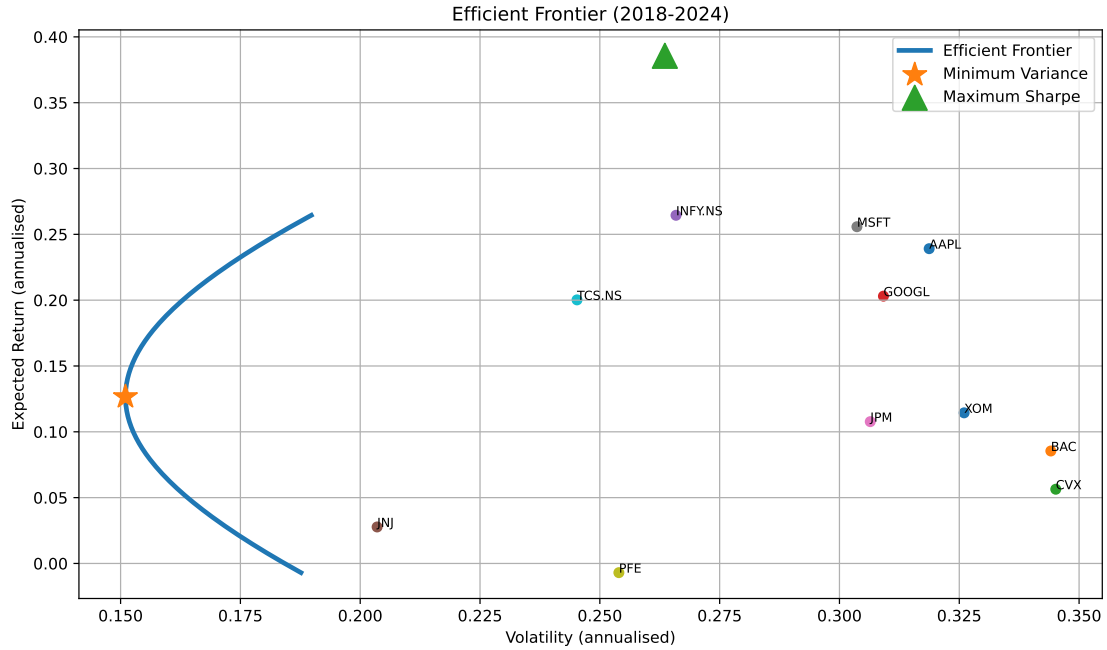


Figure 11: Efficient frontier in risk–return space. Individual assets are plotted as labelled dots. ★ = minimum-variance portfolio; ▲ = maximum-Sharpe portfolio.

Special Portfolios

Table 9: Annualised characteristics of the two key Markowitz portfolios ($r_f = 0$).

Portfolio	Ann. Return (%)	Ann. Volatility (%)	Sharpe Ratio
Minimum-Variance	12.66	15.10	0.838
Maximum-Sharpe	38.57	26.36	1.463

The minimum-variance portfolio achieves an annualised volatility of only 15.10% — well below JNJ’s stand-alone 20.35% — demonstrating the *free lunch* of diversification. The maximum-Sharpe portfolio ($SR = 1.463$) outperforms every individual asset on risk-adjusted grounds, including the best single-asset Sharpe ratio (INFY.NS: 0.995).

Portfolio Weights Analysis

Figure 12 shows the optimal weight vectors. Blue bars indicate long positions; red bars indicate short positions (permitted under the unconstrained formulation).

Minimum-Variance Weights

The minimum-variance portfolio concentrates in **JNJ** (40.6%), **TCS.NS** (23.3%), and **XOM** (17.8%) — assets with low volatility and weak correlations with the high-variance tech cluster. Short positions are taken in **CVX** (−13.0%) and **BAC** (−9.7%) to exploit their high variance in suppressing portfolio risk. The result is striking: the portfolio volatility (15.10%) is well below the lowest individual asset volatility (JNJ: 20.35%), quantifying the diversification benefit.

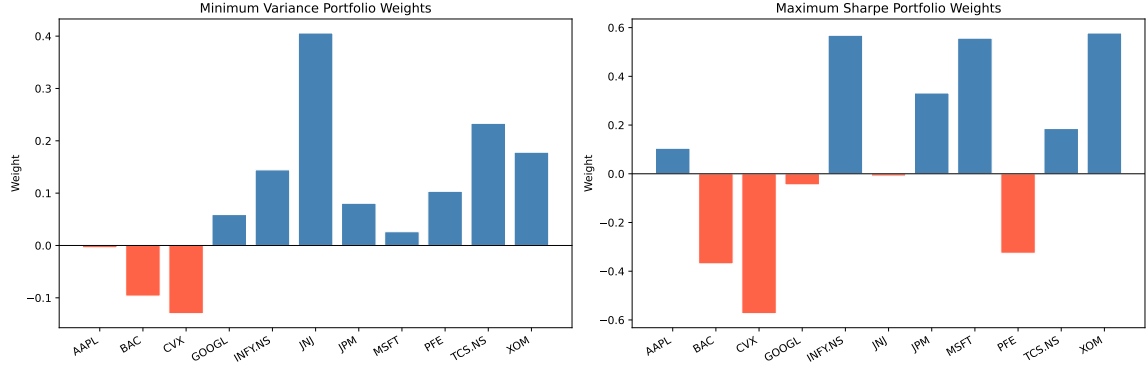


Figure 12: Optimal portfolio weights for the minimum-variance (left) and maximum-Sharpe (right) portfolios.

Maximum-Sharpe Weights

The tangency portfolio takes large long positions in **XOM** (57.8%), **INFY.NS** (56.9%), and **MSFT** (55.7%) — all assets with high individual Sharpe ratios — combined with **JPM** (33.2%) and **TCS.NS** (18.6%). It shorts heavily against **CVX** (−57.4%), **BAC** (−37.0%), and **PFE** (−32.6%), which have the lowest Sharpe ratios in the dataset.

The extreme short positions in the maximum-Sharpe portfolio (CVX −57%, BAC −37%) highlight the well-known sensitivity of unconstrained Markowitz solutions to small errors in $\hat{\mu}$. In practice, long-only constraints ($w_i \geq 0$) or shrinkage estimators are applied to produce more stable, implementable portfolios.

PCA Factor-Model vs. Full Covariance

Factor-Model Approximation

Substituting the rank- k reconstruction $\hat{\Sigma}_k = V_k \Lambda_k V_k^\top$ in place of $\hat{\Sigma}$ in (1) yields a PCA-based efficient frontier.

Explained Variance Summary

Table 10: Cumulative explained variance for the first k components.

k	1	2	3	4	5
Cum. expl. var. (%)	49.3	64.2	74.6	81.4	87.4

PC1 (49.3%) corresponds to the broad market factor. PC2 (additional 14.8%) differentiates the US tech cluster from energy and banking. Together, $k = 3$ factors explain 74.6% of the total covariance structure.

Frontier Comparison

Figure 13 overlays the efficient frontiers from the full $\hat{\Sigma}$ and from PCA approximations at $k = 2, 3, 5$.

Observations

- $k = 2$: With only 64.2% of variance explained, the rank-2 frontier deviates most from the full-covariance result, particularly in the low-volatility region, producing an optimistically

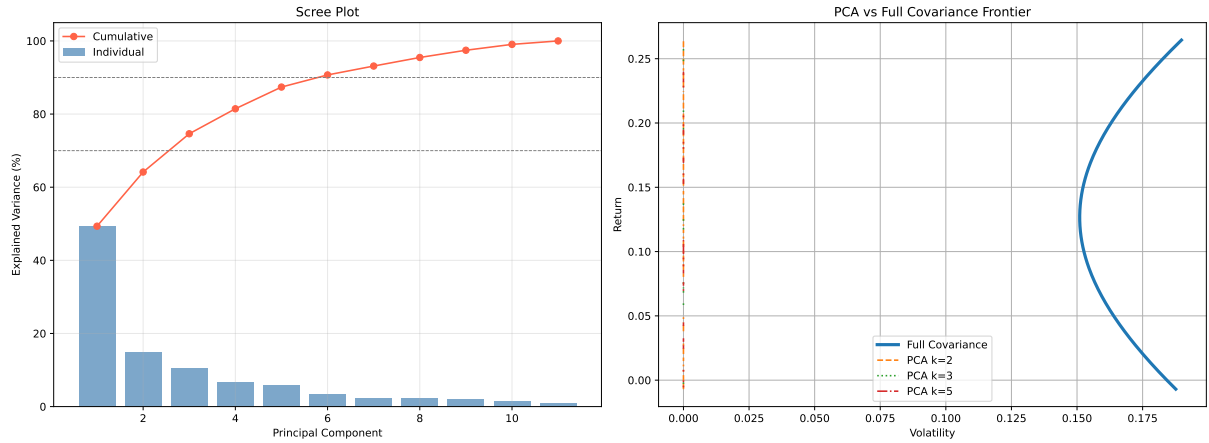


Figure 13: Left: Scree plot of individual and cumulative explained variance. Dashed grey lines mark the 70% and 90% thresholds. Right: Efficient frontiers under full covariance vs. PCA factor-model approximations at $k = 2, 3, 5$.

biased frontier.

- $k = 3$: Explaining 74.6% of variance, the three-factor frontier is substantially closer to the full-covariance result.
- $k = 5$: At 87.4% explained variance, the five-factor frontier is visually indistinguishable from the full-covariance frontier across the entire efficient range.

Insight: The convergence of the $k = 5$ frontier confirms that the 11-asset covariance matrix (66 unique entries) can be approximated by $11 \times 5 = 55$ factor loadings with negligible optimisation loss. This compression grows more valuable as the asset universe scales to hundreds of stocks.

Risk-Return Tradeoff: Discussion

Reading the Efficient Frontier

- **Every point on the efficient (upper) branch** is Pareto optimal.
- **The minimum-variance portfolio** ($\sigma = 15.10\%$) is the starting point; rational risk-minimisers hold $\mathbf{w}_{\min}$.
- **Moving right along the frontier** increases both return and volatility at a diminishing marginal rate.
- **The capital-market line** is tangent at the maximum-Sharpe portfolio $\mathbf{w}^*$; under CAPM, all rational investors hold a mix of the risk-free asset and $\mathbf{w}^*$.

Comparison with Individual Assets

All individual assets lie *to the right* of the efficient frontier. Even the best single-asset Sharpe ratio (INFY.NS: 0.995) is exceeded by the maximum-Sharpe portfolio (1.463), confirming that diversification strictly improves risk-adjusted return. The minimum-variance portfolio (15.10% vol) is less risky than any individual asset, including JNJ (20.35% vol).

Practical Limitations

1. **Estimation error.** $\hat{\boldsymbol{\mu}}$ is estimated from 1 318 daily observations; small errors produce large shifts in optimal weights. Shrinkage estimators (e.g. Ledoit–Wolf) reduce sensitivity.

2. **Time-varying volatility.** As shown in the rolling volatility analysis (Figure 5), the COVID-19 shock caused a 5–10 $\times$ spike in volatility across all assets. A rolling-window estimator would capture such regime changes.
3. **Non-normality.** Log-returns exhibit excess kurtosis and heavier tails than the Gaussian model underlying mean-variance theory. CVaR or scenario-based objectives may better capture tail risk.
4. **Short-selling.** The unconstrained solutions produce extreme short positions. Long-only constraints ($w_i \geq 0 \forall i$) produce more implementable portfolios at the cost of a tighter frontier.

Conclusion

This report derived the efficient frontier analytically via the Lagrangian closed-form solution, identified the minimum-variance ($r = 12.66\%$, $\sigma = 15.10\%$, $SR = 0.838$) and maximum-Sharpe ($r = 38.57\%$, $SR = 1.463$) portfolios, and compared the full-covariance frontier against PCA factor-model approximations at $k = 2, 3, 5$. Key findings:

1. Diversification provides a meaningful free lunch: the minimum-variance portfolio (15.10% vol) is less risky than any individual constituent.
2. The maximum-Sharpe portfolio ($SR = 1.463$) outperforms all eleven individual assets on a risk-adjusted basis.
3. Three PCA factors explain 74.6% of the covariance structure; five factors produce a frontier indistinguishable from the full-covariance result.
4. Practical deployment requires addressing estimation error, time-varying volatility, fat tails, and short-selling constraints.

References: Deisenroth, Faisal & Ong — Mathematics for Machine Learning. Data: Yahoo Finance via yfinance. Analysis period: 2018-01-01 to 2024-01-01 ($T = 1318$ daily observations).